

2022 Ph H2 Q10

A ray of monochromatic light is incident on a glass block. The question involves calculating angles and applying Snell’s Law.

(a)
Calculate the refractive index of the glass

Given:

- Angle of incidence $i = 36.0^\circ$
- Angle of refraction $r = 22.0^\circ$

Use Snell’s Law:

$$n = \frac{\sin i}{\sin r} = \frac{\sin(36.0^\circ)}{\sin(22.0^\circ)} = \frac{0.5878}{0.3746} = 1.57$$

(b)
State what happens to the speed of the light as it enters the glass

✔ The light **slows down** as it enters the glass, because:

- Glass has a **higher refractive index** than air
- Light travels **more slowly in denser optical media**

(c)
Calculate the critical angle at the glass-air boundary

Use:

$$\sin c = \frac{1}{n} = \frac{1}{1.57} = 0.6369 \Rightarrow c = \arcsin(0.6369) = 39.6^\circ$$

(d)
State what happens to the refractive index if light of a different wavelength is used

✔ The refractive index depends on **wavelength**:

- **Shorter wavelengths** (e.g. blue light) are refracted **more strongly**
- So, using a different wavelength **changes** the refractive index value.

🧠 Revision Tips

- Snell’s Law: $n = \frac{\sin i}{\sin r}$; always use angles **measured from the normal**
- Refractive index is **greater than 1** for light slowing down
- **Critical angle** occurs when light travels from a denser to a rarer medium
- Be ready to explain physical consequences (e.g. speed, direction, or colour dependence) of changing medium or wavelength